Data-Driven Optimization in Bottle Manufacturing

A Final report for the BDM capstone Project

Submitted by

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Content

1. Executive Summary

SP Enterprises is a plastic manufacturing company that deals with various raw materials such as HDPE, PET, PP, PVC, and HIPS. The company has recently faced several operational challenges like **inconsistent raw material usage**, **unpredictable customer demand**, **inefficient storage space, and poor planning in procurement**. These problems were affecting their supply chain, production planning, and overall efficiency. To address these concerns, a structured data analysis was conducted using multiple analytical methods to understand patterns, identify key issues, and provide practical solutions.

The first step in the analysis was to study the **purchase trends of raw materials** over five months. This showed that while PET had a stable usage pattern, materials like HDPE experienced sharp fluctuations. The next step was **correlation analysis**, where we checked if there was any relationship between customer bottle orders and the purchase of specific raw materials. We found a moderate negative correlation (-0.35), which suggests that in some cases, an increase in raw material purchases did not always match customer demand — highlighting inefficiencies in procurement. To prioritize which raw materials and clients to focus on, **ABC analysis** was used. This helped classify materials and customers into three categories: A (most important), B (moderately important), and C (least important). For raw materials, it showed that a small number of items contributed to a large portion of total costs. Similarly, client analysis showed that only a few clients accounted for most of the bottle orders. To further help in decision-making, a **3-month moving average forecast** was used to predict future raw material demand. This forecasting method gave a smoother picture of future needs and can guide monthly purchasing plans to avoid over- or under-ordering.

Based on the analysis, several valuable findings emerged. These insights directly relate to the company's earlier problems. As a result, a set of **recommendations** has been given to SP Enterprises, including maintaining optimal stock levels for Class A materials, building strong relationships with key clients, reducing focus on low-value items, improving purchase planning based on forecasts, and using storage space more effectively. If these recommendations are followed, SP Enterprises can significantly improve its operations, save costs, and better meet customer demand.

2. Detailed Explanation of Analysis Methods

2.1 Trend Analysis

I am using trend analysis to understand the changing patterns in raw material procurement and bottle demand over a 5-month period (January to May 2025) at **SP Enterprises**, a bottle manufacturing unit in Bhagwanpur, Haridwar. Given the company's small scale, with limited storage capacity and variable client orders, identifying procurement and demand trends is essential for **optimizing operations, avoiding stockouts**, and ensuring smoother production cycles.

Trend analysis helps detect seasonality , procurement delays, or overstocking. This understanding supports better alignment between raw material purchases and actual client demand.

Steps to Perform:

- **Data Preprocessing:**
  - Aggregate bottle demand and raw material procurement by month.
  - Organize data using PivotTables in Excel (e.g., total bottles ordered per month, total kg of HDPE per month).

- **Visualize Trends:**
  - Plot line charts for each raw material and total bottle demand.
  - Use different colored lines for each material to compare fluctuations.

- **Smooth Trends:**
  - Apply **3-point Moving Average** to smooth monthly variations and identify the long-term trend.

- **Identify Peaks and Dips:**

- Annotate months with significant demand spikes (e.g., January bottle spike) or drops (e.g., April procurement dip).

Expected Outcome:

This analysis will help SP Enterprises recognize consistent monthly fluctuations in both procurement and client demand. It enables pre-planning of raw material needs and reduces risks of over/under-purchasing. It will also allow the company to prepare for high-demand months like January and May, optimizing inventory and cash flow.

2.2 Correlation Analysis

Correlation analysis measures how two variables move together. In this project, we assess:

- The relationship between **bottle demand** and **raw material procurement**

To see the present **raw material procurement strategy**, we will check if the amount of raw material purchased matches the number of bottles ordered. If both go up or down together (positive correlation), it means planning is good. If not, it shows that raw materials may be bought without knowing the actual demand, which leads to waste or shortage. We will also see **demand and supply imbalance** by looking at the link between raw material purchase and how much is produced. If there is little or no connection, it may mean delays in production or not enough supply when needed. This shows a mismatch between what is needed and what is made.

Steps to Perform:

- **Data Preprocessing:**
  - Summarize monthly totals for bottles ordered, raw materials purchased, and their prices.
- **Apply Correlation Formula:**
  - Use Excel's =CORREL(array1, array2) to calculate Correlation.
- **Interpret Results:**

- Correlation close to +1: strong positive relationship
- Correlation close to -1: strong negative relationship
- Near 0: weak/no relationship

Expected Outcome:

It will help us in analyzing if company is buying materials the right way and if production is matching demand or not. A **strong positive correlation** between raw material procurement and bottle demand would indicate well-aligned planning, while a **weak or negative correlation** may reveal inefficient procurement practices not based on actual demand. A **low correlation** between raw material availability and production output may point to production delays or supply mismatches, highlighting demand-supply imbalance and possible bottlenecks.

2.3 ABC Segmentation

ABC segmentation is a technique used to group raw materials and clients based on their importance usually measured by their contribution to total cost, usage, or order volume. For SP Enterprises, this method can help identify which materials and clients should be given the most attention.

For example, a material like **HDPE**, if it turns out to be costly and widely used, may fall under **Class A**, while a less used material like **PVC** may fall under **Class C**. Similarly, clients who place large or regular orders could be considered **Class A clients**.

Steps to Perform:

- **Data Preprocessing:**

  - Calculate the total quantity and total cost for each raw material (e.g., Total Price = Quantity * Price per kg).
  - Calculate total bottles ordered per client.

- **Calculate Cumulative Contribution:**

- Sort materials/clients in descending order of total value or volume.
- Calculate % contribution to overall total and cumulative percentage.

- **Classify into A, B, C:**

- **Class A:** Top 70-80% contribution
- **Class B:** Next 15-20%
- **Class C:** Remaining low-impact materials/clients

- **Visualize Using Pie/Bar Charts:**

- Using Excel bar to make Pareto charts to visualize classification impact.

Expected Outcome:

This segmentation enables SP Enterprises to adopt differentiated strategies. Identify high-value materials (Class A) that need close monitoring and regular procurement**.** Spot mid-importance items (Class B) where inventory levels can be optimized**.** Recognize low-impact materials (Class C) that may allow for reduced stock or less frequent purchases. This approach will support **better resource planning**, **cost control**, and **customer satisfaction** by helping the company focus on what matters most.

2.4 Forecasting Using Moving Averages

Forecasting helps predict future demand for bottles or raw material needs based on past trends. At SP Enterprises, forecasting is critical for planning purchases, production, and workforce allocation. Using **moving averages** in Excel, we can project next month's bottle demand or material quantity needs, enabling proactive decision-making.

Steps to Perform:

- **Prepare the Data:**

- Create two columns: Month Index (e.g., Jan = 1, Feb = 2, ...) and the target variable (e.g., total bottles or HDPE kg).

- **Calculate Moving Averages in Excel:**

- Using Excel to calculate 3-month moving average:

  For example:

  =(B1+B2+B3)/3 for a 3-month moving average

- **Extend the Series:**

Use the latest moving average values to forecast demand for June, July, and August 2025.

- **Plot the Trend:**

  Create a line chart showing actual values and moving averages to visualize the trend.

Expected Outcome:

SP Enterprises can predict raw material requirements for June, July, August 2025 based on past trends, preventing procurement delays and overstocking. Forecasting bottle demand will also guide production scheduling, ensuring timely client delivery and improving overall efficiency. Overall, this method supports **better planning and operational efficiency** by using past data to anticipate future needs.

3. Results and Findings

3.1 Trend Analysis of raw material purchase over five months

- SP Enterprises is facing problems in the way it buys raw materials. A chart showing the monthly purchase of five materials—HDPE, HIPS, PET, PP, and PVC—shows that the buying pattern is not stable. For example, the quantity of **HDPE** keeps going up and down every month. It was high in January, went even higher in March, then dropped in April, and rose again in May. This means the company is not planning its purchases properly and may be buying too much or too little without knowing the actual need.
- **HIPS** also shows a sudden jump in February after a low purchase in January. This could mean that the company did not expect the demand or didn’t plan well. On the other hand, **PET** shows a steady and slow increase from January to May, which is a good sign. It means that

PET is being purchased in a planned and regular way. PP is mostly stable with small changes, which is also okay. **PVC**, however, shows a slow decrease in quantity, which could mean that the company is either using less of it or not managing the purchase correctly.

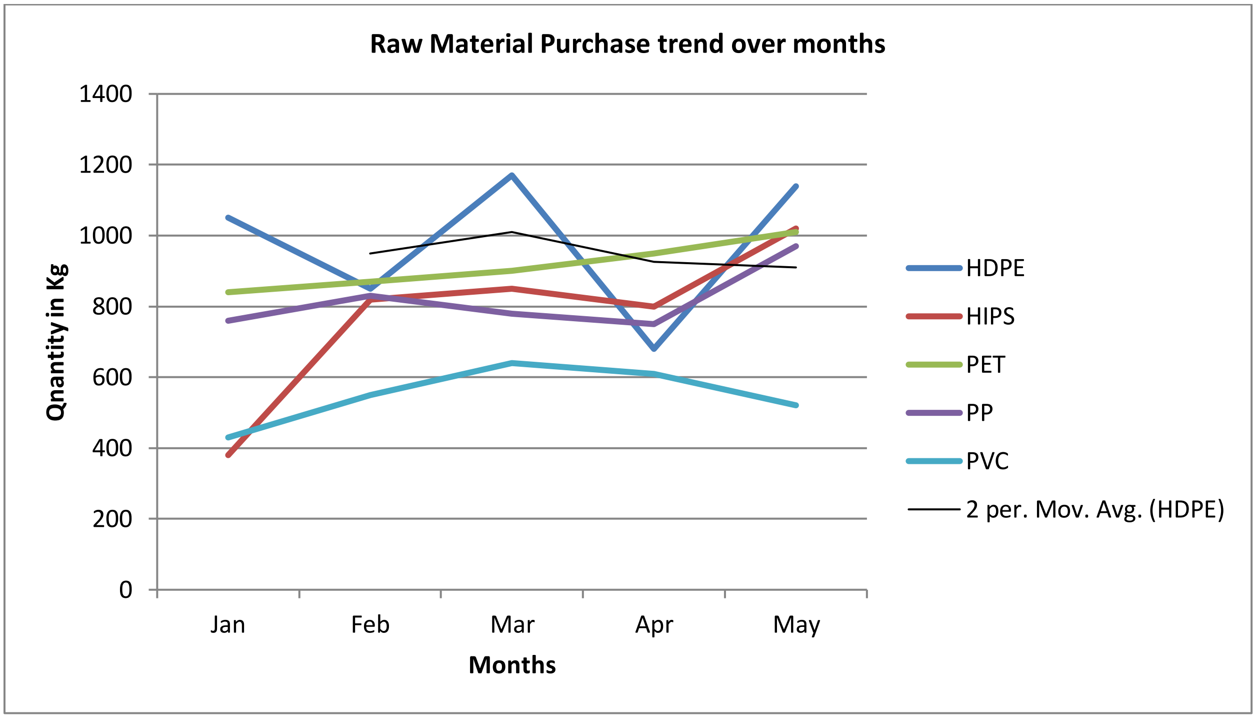

Figure 1 Monthly Raw Material Procurement Trend (Jan–May 2025)

- These patterns give us some useful ideas. Materials like HDPE and HIPS are not being managed well. There is no clear plan or method being followed, which causes waste or shortages. The difference in trends between materials also shows that the company is not thinking of all materials together while planning. Even the line used to smooth the HDPE data (moving average) doesn't match the actual numbers, which shows that the method used to guess future needs isn't working well.
- So, SP Enterprises needs to improve its way of buying raw materials. It should start using better methods to guess how much material will be needed in the future. It should also plan all material purchases together, not separately. This will help avoid buying too much or too little and save money. Good planning will also make sure the production work doesn't stop due to missing materials. In short, a better and smarter buying system will solve the problem of poor procurement and help the company work more smoothly. To better understand whether these

raw material purchases are demand-driven or not, a correlation analysis was performed between raw material purchased and number of bottles ordered.

3.2 Correlation Analysis between raw material purchased and number of bottles ordered by clients

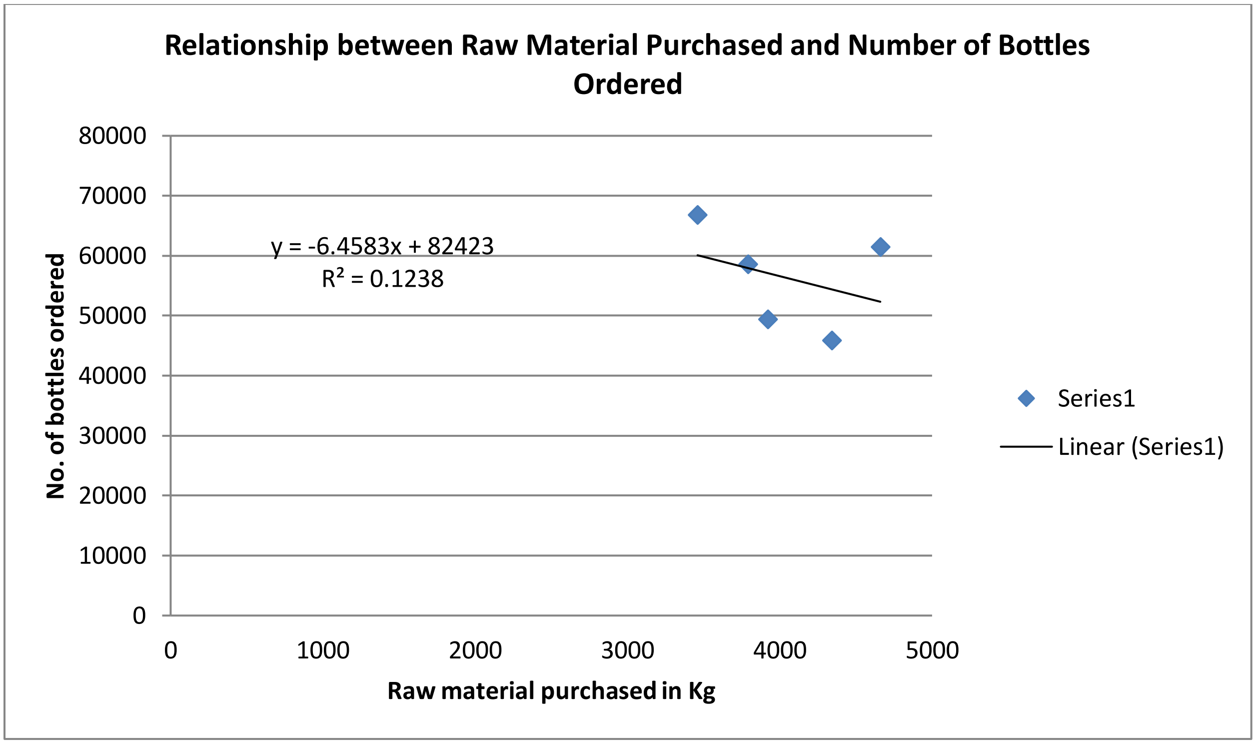

Figure 2 Correlation Scatter Plot – Raw Material vs Bottle Orders

- The scatter plot shows the relationship between raw material purchased (in Kg) and the number of bottles ordered over five months. From the graph, we can see that there is **no strong link** between how much raw material was bought and how many bottles were ordered. The data points are scattered, and the line in the graph slightly slopes down. This means that even when more raw material was purchased, bottle orders didn't always increase. The **correlation value is -0.35**, which shows a **moderate negative relationship** between the two — as raw material increases, bottle orders slightly decrease. The R^2 value (which tells us how closely the two things are connected) is very low — only **0.123** — which shows a **weak negative correlation**.
- This result clearly supports the first problem: **Inefficient Raw Material Procurement Strategy**. If raw material purchases were properly planned based on actual bottle orders or

expected demand, we would see a stronger and more positive pattern in the graph. But here, since there's no clear connection, it looks like raw materials are being bought without checking how many bottles are actually needed. This kind of random or poorly planned buying leads to wastage, overstocking, or sometimes even shortage of required materials.

- The chart also points toward **Demand-Supply Imbalance and Production Bottlenecks**. If materials are not bought in the right quantity at the right time, production may suffer. Sometimes there might be too much raw material lying unused, and other times there might not be enough to meet urgent orders. This mismatch between purchase and demand creates problems in smooth production and causes delays or stoppages in bottle-making.
- Finally, when raw materials are over-purchased without matching production needs, it leads to **space problems** in the warehouse or storage areas. If materials are unused for long periods, they occupy space that could have been used for other important items or ongoing production. This makes daily operations harder and less efficient. So, this scatter plot clearly shows why SP Enterprises needs to improve its planning of raw material purchases to fix these three problems — poor buying strategy, mismatch between supply and demand and storage issues.

3.3.1 ABC Segmentation of raw materials

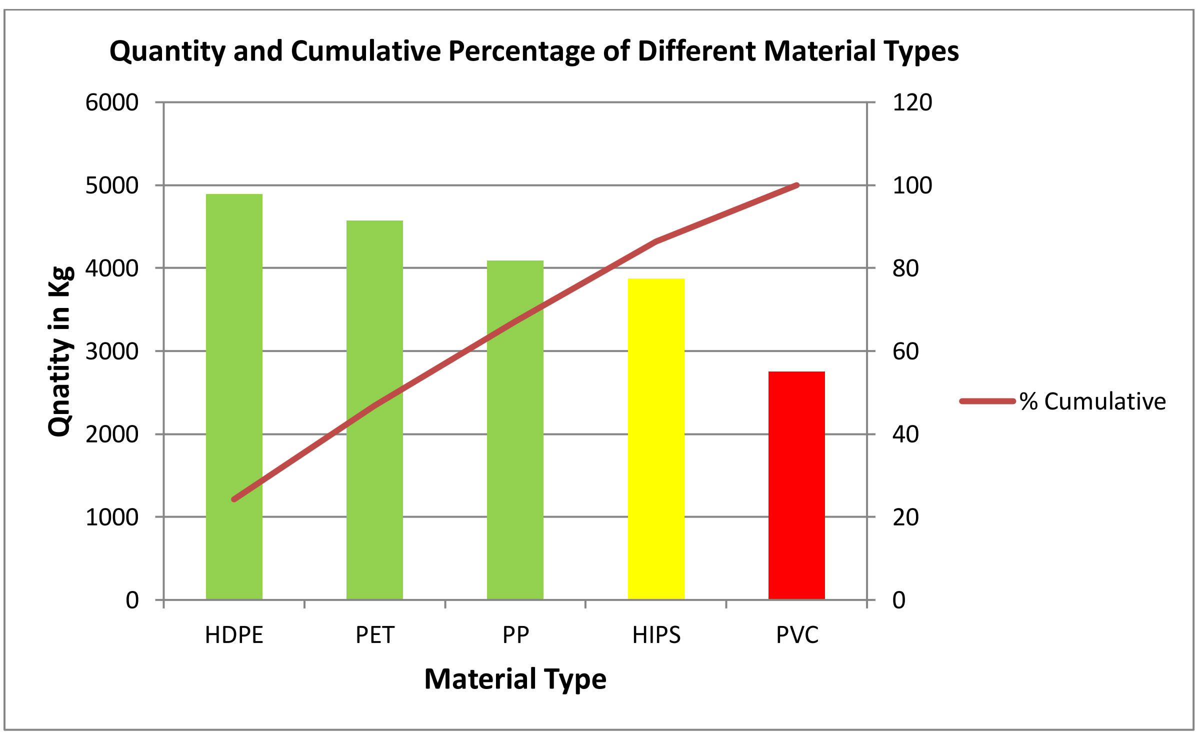

Figure 3 ABC Segmentation of Raw Materials based on Cumulative Procurement Contribution

- The ABC analysis chart categorizes raw materials—HDPE, PET, PP, HIPS, and PVC—based on their total procurement quantity and cumulative contribution. From the chart, HDPE and PET are shown in green and make up the top two contributors to raw material usage, falling into **Category A**. These are high-priority items due to their high consumption and cost. PP, also in green, borders between A and B. HIPS is in **Category B** (yellow), while PVC, with the lowest quantity and impact, falls into **Category C** (red). This classification helps SP Enterprises understand which materials drive the most value and require the most attention.
- This analysis directly helps address the **problem of inefficient procurement**. Previously, the company may have been giving equal importance to all materials. However, by identifying HDPE and PET as A-class items, the company can now prioritize their procurement, quality checks, and stock levels. These are the materials that must be monitored closely for price changes and availability, as any delay or error in managing them could directly affect production. On the other hand, C-class materials like PVC can be ordered in smaller, less frequent batches, saving cost and storage space.
- ABC analysis also helps tackle the **space constraint problem**. Warehousing space is often wasted when all items are stored equally, regardless of usage frequency. With this classification, SP Enterprises can optimize their storage: high-priority A items like HDPE and PET can be stored closer to the production floor in larger quantities, while low-priority C items like PVC can be stored in secondary locations or ordered only when needed. This approach reduces clutter, improves material flow, and ensures that critical materials are always within reach, making the entire operation smoother.
- Lastly, this analysis supports solving the **demand-supply mismatch and production delays**. By focusing on the A and B materials, the company ensures that production-critical inputs are never out of stock, preventing unnecessary stoppages. For instance, if HDPE or PET runs out, the impact is far more severe than if PVC is missing. So now, the procurement team can build vendor relationships and reorder plans for A materials well in advance, improving the reliability of the supply chain. Overall, ABC analysis is a strategic tool that enables SP Enterprises to align procurement priorities with actual business needs and optimize operations effectively.

3.3.2 ABC Segmentation of clients

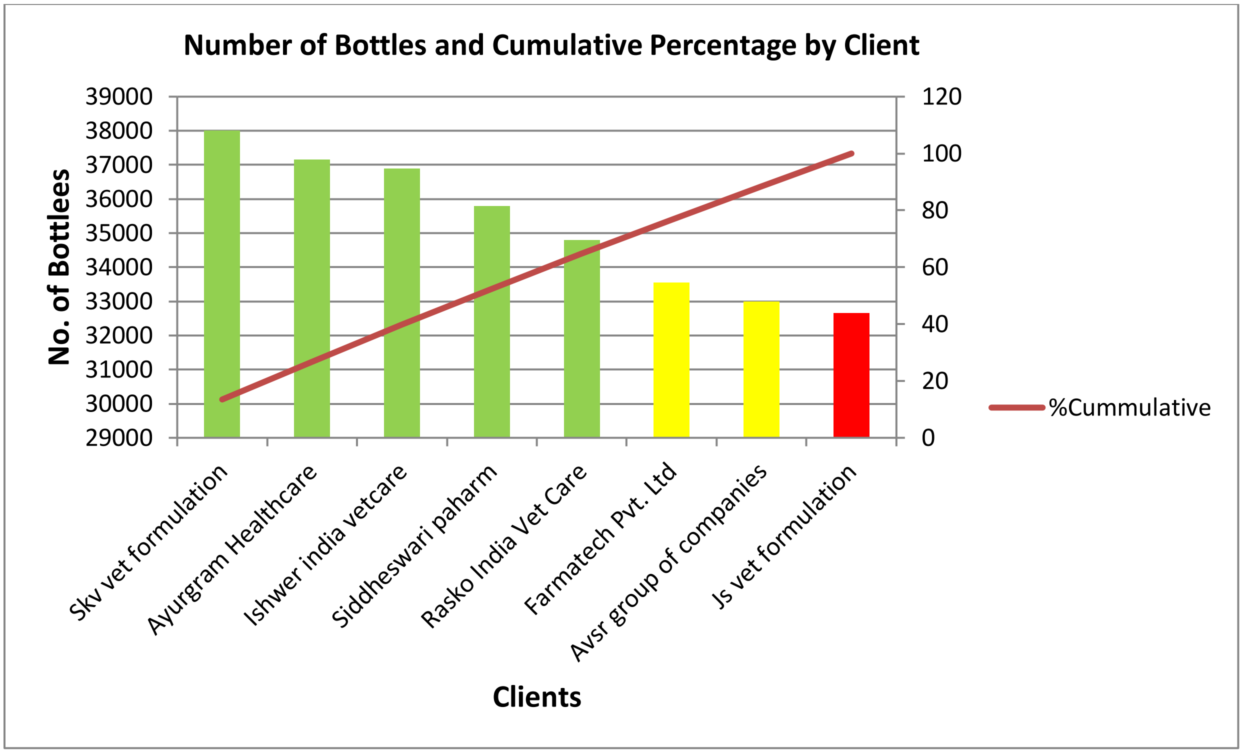

Figure 4 ABC Segmentation of Clients based on Bottle Order Volumes

- The client **ABC segmentation** chart categorizes customers based on the volume of bottles ordered, helping SP Enterprises identify which clients contribute most significantly to overall sales. From the visualization, it's evident that a small number of clients are responsible for a large portion of the total orders. Clients such as Skyvet Formulation, Ayurgram Healthcare, Ishwer India Vetcare, Siddheswari Paham, Rasko India, and Vet Care fall into **Category A.** These clients are highlighted in green and represent the **top 70%** of total orders. Their consistent and high-volume purchases make them critical to the business's revenue flow, and retaining these clients is essential for the company's long-term stability.
- Previously, SP Enterprises might have been allocating equal attention to all clients regardless of their contribution to sales, which is inefficient. By using ABC segmentation, the company can now adopt a more focused strategy. High-priority clients in Category A should be given preferential treatment—personalized service, priority order fulfillment, and perhaps exclusive offers—to encourage continued loyalty and even higher engagement. These clients should also be included in feedback loops for product improvements or trials for new packaging solutions, as their satisfaction directly impacts overall performance.

- **Category B** clients, shown in yellow, include companies like Farmatech Pvt. Ltd. and Avsr Group of Companies. While they don't contribute as much as Category A clients, they are still important and hold growth potential. These clients can be targeted with moderate relationship management efforts—perhaps occasional check-ins, promotional offers, or loyalty programs—to gradually increase their order volume. By strategically engaging this segment, SP Enterprises can move some of these clients into Category A over time, thus improving overall sales diversity and reducing dependency on just a few key clients.
- **Category C** clients, such as Js Vet Formulation, contribute very little to overall sales and are represented in red. While these clients are still important, they require fewer resources and can be handled using more standardized processes. Understanding this distribution helps SP Enterprises optimize how much time and energy they spend across different clients. For example, logistics and sales teams can prioritize A clients during inventory shortages or delivery scheduling, ensuring that the most valuable relationships are always served first. Overall, ABC segmentation transforms client management from a one-size-fits-all approach into a data-driven, prioritized strategy aligned with actual business impact.

3.4 Demand Forecasting of Number of Bottles by 3-month moving average forecast

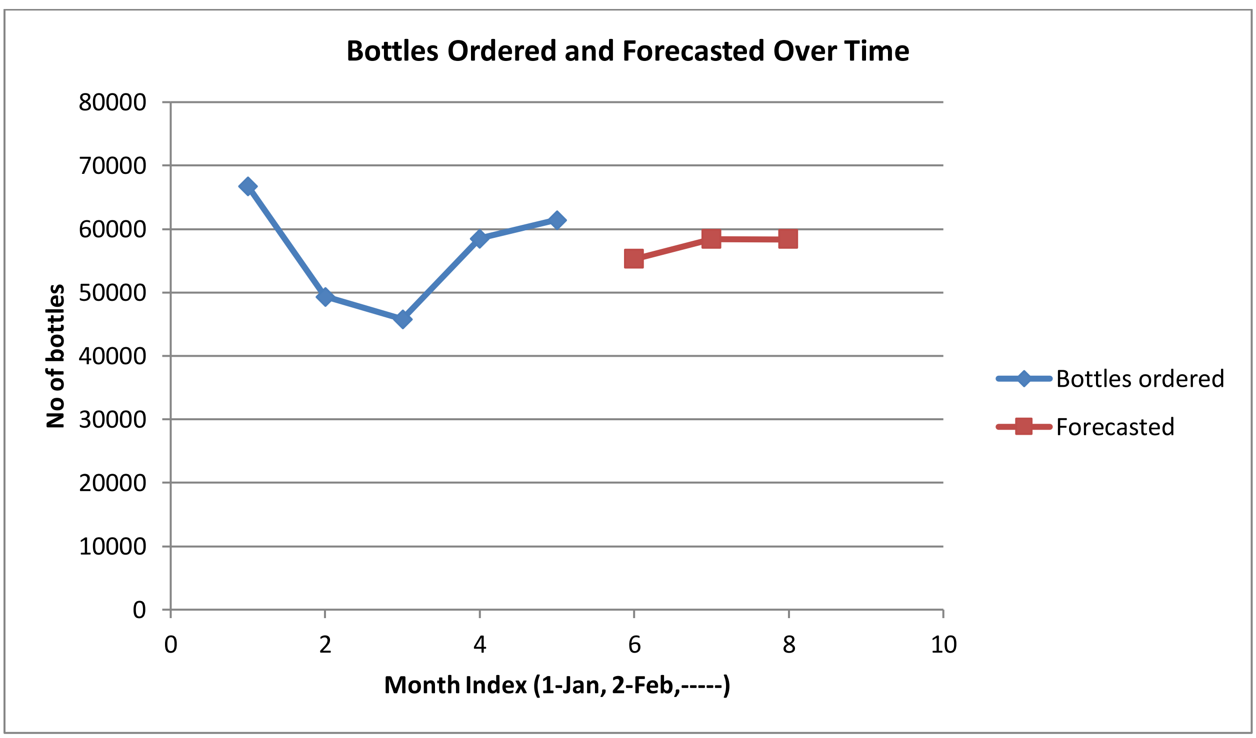

Figure 5 Forecast of Bottle Orders (June–August) Using 3-Month Moving Average

- The chart shows the monthly demand for bottles from January to May and a forecast for June, July, and August using a 3-month moving average method. In the beginning, bottle orders were high in January (around 66,750), but they dropped sharply in February and March. After March, there was a slow rise in demand in April and May. To understand what might happen next, we used a simple technique called the 3-month moving average, which takes the average of the past three months and uses it to predict the future. According to the chart, the forecasted demand for bottles is expected to be steady around 55,000 to 59,000 bottles in the coming months. This gives us a helpful idea of what to expect and helps in preparing better.
- This kind of analysis is important for **solving the problem of inefficient raw material procurement**. In the earlier trend analysis, we saw that raw materials were purchased without a clear link to bottle demand. For example, even when bottle orders were low, some materials like HDPE were bought in high quantities. This mismatch could lead to wastage, overstocking, or shortage of key materials. But now, with a proper forecast of bottle demand, SP Enterprises can plan better. They can look at how much raw material will actually be needed based on predicted bottle orders and only buy the amount that's required. This ensures smarter procurement and reduces unnecessary costs.
- Another problem this forecast addresses is the **demand-supply imbalance and production delays**. If the company doesn't know how much demand is coming, they may either produce too much or too little. This creates stress on the production line and workers, and may even delay delivery to customers. But by forecasting the next three months, the company can start preparing in advance. If demand is likely to increase, they can arrange for more staff, more materials, and better scheduling. If demand is likely to drop, they can slow down production, save energy, and reduce overtime. This makes the entire production process more efficient and reduces the chances of any bottlenecks.
- Finally, this forecast is also useful for solving the issue of **space constraints**. When too many raw materials or finished goods are stored at once, it causes space problems in the factory or warehouse. Often, this happens because companies overstock due to poor planning. But now, using this demand forecast, SP Enterprises can manage space better. They will know how much to produce and store, reducing the risk of overflow. It also saves money because they don't have to rent or build extra storage spaces. So, forecasting is not just about numbers—it

helps with better decisions in procurement, production, and storage, and directly supports the company in fixing its key challenges.

4. Interpretation of Results

- From the monthly data of raw materials, we can see that **PET purchases increased gradually from January to May**, indicating a rising and consistent demand. This pattern suggests PET is a key material used in the production process and its importance is growing. On the other hand, **HDPE showed unpredictable behavior**, with a sharp rise in March followed by a sudden drop in April. This kind of fluctuation means HDPE might be used in specialized or seasonal products, or its demand is influenced by specific external factors.
- Materials like **HIPS and PP had moderate but noticeable changes each month**, showing that they are used fairly regularly. They neither had steep increases nor large drops, which means their usage is somewhat stable. **PVC stayed consistently low across all five months**, with minimal change in quantity ordered, indicating it's either used very little or only for specific niche purposes.
- When analyzing client orders, we observed that **only a small number of clients were responsible for the majority of bottle orders**. This means the business depends heavily on a few key clients for revenue and should make special efforts to retain these clients through personalized service, fast delivery, and strong communication.
- In contrast, **most other clients placed only small or occasional orders**, contributing less to the total. While they are still important, they do not significantly impact the company's monthly revenue. However, there's an opportunity to convert some of these clients into bigger, more regular buyers with better engagement or offers.
- We also studied the relationship between raw materials, and **found that PET and PP often showed similar ordering trends**, which may mean they are used in the same or related types of products. This can be used to plan better inventory — if demand for one rises, the other is likely to follow.
- On the other hand, materials like **HDPE and PVC did not show any clear relationship**, meaning their usage patterns are independent. This means there's no benefit in jointly managing their orders or stock, and they should be handled separately to avoid overstocking or wastage.

- Using ABC classification, we divided raw materials based on their total usage. **Class A materials like PET and HDPE are the most used and most important**, so they should always be in stock, closely monitored, and ordered in larger quantities to avoid stockouts.
- **HIPS and PP fall into Class B**, meaning they are moderately used. They still require monitoring but not as frequently or urgently as Class A materials. **PVC, being the least used, is in Class C**, so it can be ordered less frequently, and kept in limited quantities.
- The same ABC method was applied to clients. **Class A clients placed large and regular orders**, making them highly valuable to the company. These clients should be prioritized in every business process — from order processing to delivery and after-sales service.
- **Class B clients gave medium-sized orders**, and while they are not as critical as Class A, they still have potential. Regular follow-ups or offers might help them move to the A category. **Class C clients placed small or irregular orders**, so they can be managed with general communication and automated systems.
- To prepare for the future, we used forecasting to predict material needs for June to August. **PET demand is expected to continue growing**, so the company must ensure sufficient stock in advance to meet the rising requirement without delays in production.
- Though **HDPE's trend was not smooth**, forecasts suggest it will still be needed in significant amounts, just not in a predictable way. This means the company should not overstock it, but should be ready to purchase quickly when demand rises.
- Client order forecasting showed that **Class A clients will likely continue placing large or growing orders in the next few months**. This gives the company confidence to plan production ahead of time and reduce lead times, improving customer satisfaction.
- Overall, this analysis gave clear insights into how raw materials are used, which clients are most valuable, how materials relate to each other, and what the future might look like. All these findings help the business make smarter decisions, reduce waste, avoid shortages, and run operations more smoothly and efficiently.

5. Recommendation to Business

1. Focus more on PET

PET is the only raw material that showed a steady increase in purchases from January to May. This means it's being used more and more. The company should always keep enough PET in stock to avoid delays in production. It’s also a good idea to negotiate long-term deals with suppliers for PET, as this might lower the cost and ensure a constant supply.

2. Keep a flexible plan for HDPE

HDPE usage is not stable — it increases one month and decreases the next. This makes it risky to place the same order every time. Instead, the company should review recent usage before ordering HDPE each month. This way, they don’t overstock or run out unexpectedly.

3. Build strong relationships with top clients

From the client ABC analysis, we saw that a few top clients (Class A) are responsible for the majority of the business. Losing even one of them could cause a big drop in sales. The company should maintain good contact with them, offer small discounts or loyalty rewards, and try to understand their future needs.

4. Plan raw material purchases based on usage patterns

The correlation analysis showed that some materials like PET and PP are often used together. So if PP is being used, chances are PET is also needed. The company should think about ordering these together and storing them near each other to make handling easier and faster.

5. Don’t overstock Class C materials

Class C raw materials, such as PVC, are rarely used. Keeping a large amount of such items takes up space and blocks money that could be better used elsewhere. The company should buy these only when there's a clear need, maybe even after getting a customer order.

6. Use ABC classification to manage inventory better

ABC analysis can help in stock management. Class A items need strict tracking and fast reordering because they are used the most. Class B items can be checked regularly, while Class C items need very little attention. This method saves time and keeps the warehouse organized.

7. Give attention to Class B clients too

While the business must take care of its top clients, mid-level clients also contribute a fair share. They have the potential to become top clients with better service or offers. Reaching out to them more often, asking for feedback, and offering occasional promotions can increase their order size.

8. Use forecasts to prepare better

The 3-month moving average forecast helps predict how much of each raw material might be needed in the future. These predictions can guide the purchase department to plan better and avoid both shortages and excess stock. Forecasting should be done regularly, especially for Class A materials.

9. Train the team to read trends regularly

The team managing raw materials and client relations should be trained to read and understand trend charts. By checking monthly patterns, they can identify if any material is being overused or underused and act in time. This proactive approach can reduce mistakes and improve planning.

10. Explore why some materials or clients have low activity

It's important to not ignore the low-performing raw materials or clients. Maybe PVC is not used because of a change in production methods, or a client is inactive due to late deliveries. Finding out the reasons can help the company improve its services or even bring in more orders from those sources.